Iris Flower Classification

Exploratory analysis and species model comparison

PROJECT SUMMARY

This project classifies iris flowers into Setosa, Versicolor, and Virginica from sepal and petal measurements using scikit-learn data built into Python.

DATA AND SCOPE

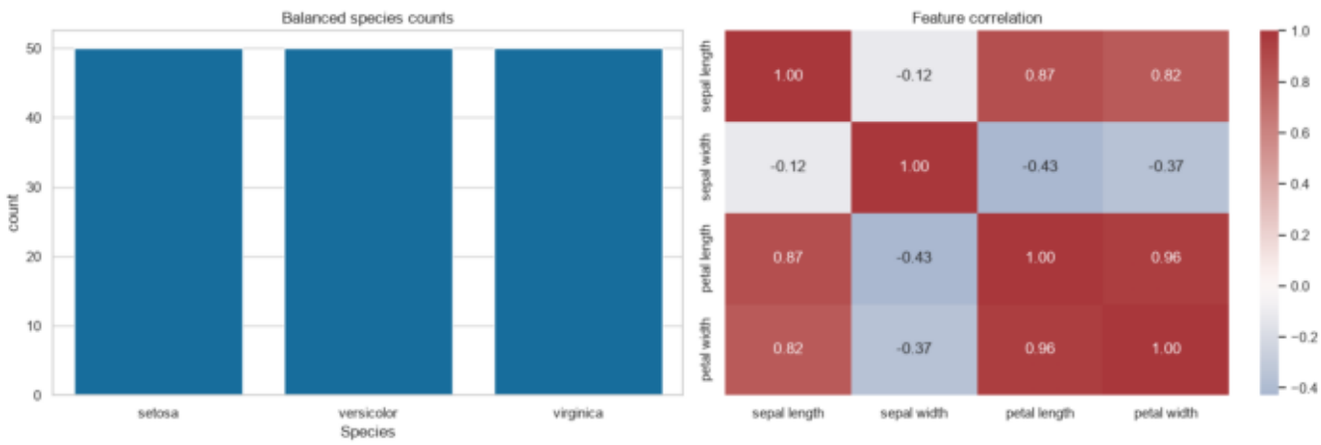
150 observations, four numeric measurements, and three balanced species classes. No external dataset download is needed.

ANALYSIS APPROACH

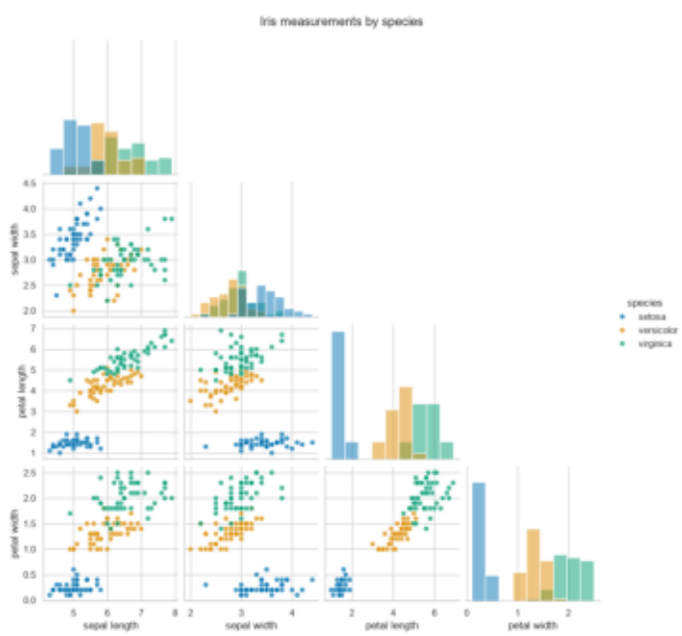
Reviewed shape, data types, missingness, descriptive statistics, species counts, pairwise feature plots, boxplots, and feature correlations. Compared standardized Logistic Regression and K-Nearest Neighbours on a stratified 80/20 split. Used five-fold cross-validation on the training partition as a stability check.

RESULTS AND EVIDENCE

Both tested models reached 0.933 held-out accuracy. Both had mean training-fold cross-validation accuracy of 0.958 (SD 0.026). Logistic Regression is listed first as the tied model selected by the notebook. Petal length and petal width show the clearest species separation in the plots; sepal measurements overlap more.



Selected notebook visualization



Selected notebook visualization

INTERPRETATION AND REPRODUCTION

Limitations and responsible use

Only 30 flowers are in the holdout set, so one classification changes accuracy by about 3.3 percentage points. This clean benchmark does not represent measurement noise or deployment conditions.

Reproduce the analysis

Open iris-flower-prediction-mlnotebook.ipynb in Jupyter and run all cells. The dataset loads from scikit-learn.

The notebook includes the complete EDA, charts, modeling code, metric tables, and cell outputs.

Interpret model scores in the context of the dataset and holdout design. Observational patterns and predictive associations do not establish causal effects.